

# FinalProject

2025-03-13

```
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(readr)
library(factoextra)

## Loading required package: ggplot2
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa

library(ggplot2)
library(cluster)

airbnb <- read_csv("listings.csv")

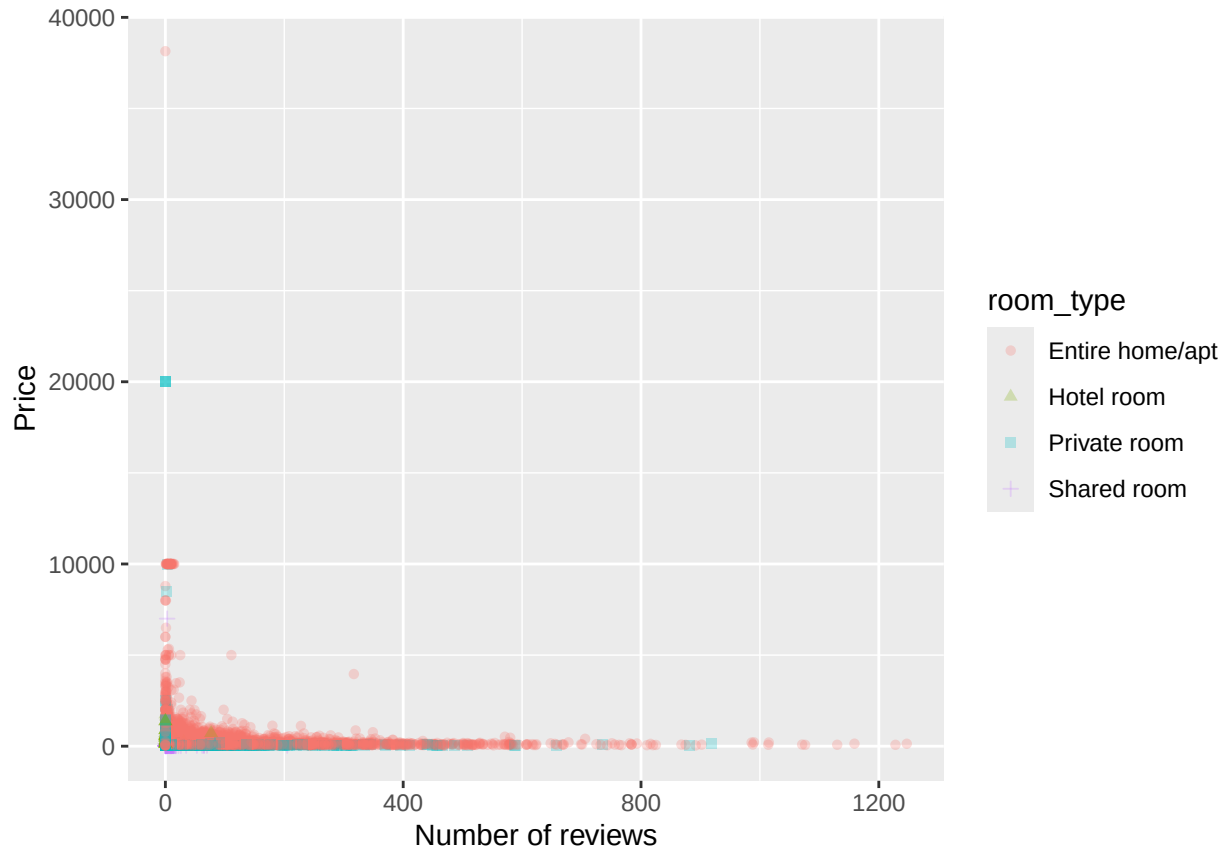
## Rows: 15500 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (3): name, host_name, room_type
## dbl   (12): id, host_id, neighbourhood, latitude, longitude, price, minimum_n...
## lgl   (2): neighbourhood_group, license
## date  (1): last_review
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
# Round data into 2 decimals place for better computation and readability
airbnb |>
  select(price, minimum_nights, number_of_reviews) |>
  cor(use = "pairwise.complete.obs") |>
  round(2)

##           price minimum_nights number_of_reviews
## price           1.00          -0.02           -0.06
## minimum_nights  -0.02           1.00           -0.06
## number_of_reviews -0.06          -0.06           1.00

ggplot(airbnb, aes(number_of_reviews, price, color = room_type, shape = room_type)) +
  geom_point(alpha = 0.25) +
```

```
xlab("Number of reviews") +
ylab("Price")
```

```
## Warning: Removed 4126 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
airbnb_2cols <- airbnb[, c("price", "number_of_reviews")]

airbnb[, c("price", "number_of_reviews")] = scale(airbnb[, c("price", "number_of_reviews")])

airbnb_na_omit <- na.omit(airbnb_2cols)

set.seed(123)
km.out <- kmeans(airbnb_na_omit, centers = 3, nstart = 20)
set.seed(345)
km.out2 <- kmeans(airbnb_na_omit, centers = 3, nstart = 20)

airbnb_cols <- airbnb[, c("minimum_nights", "neighbourhood", "price", "number_of_reviews")]

airbnb_cols[, c("minimum_nights", "neighbourhood", "price", "number_of_reviews")] = scale(airbnb_cols[, c(

airbnb_na <- na.omit(airbnb_cols)
```

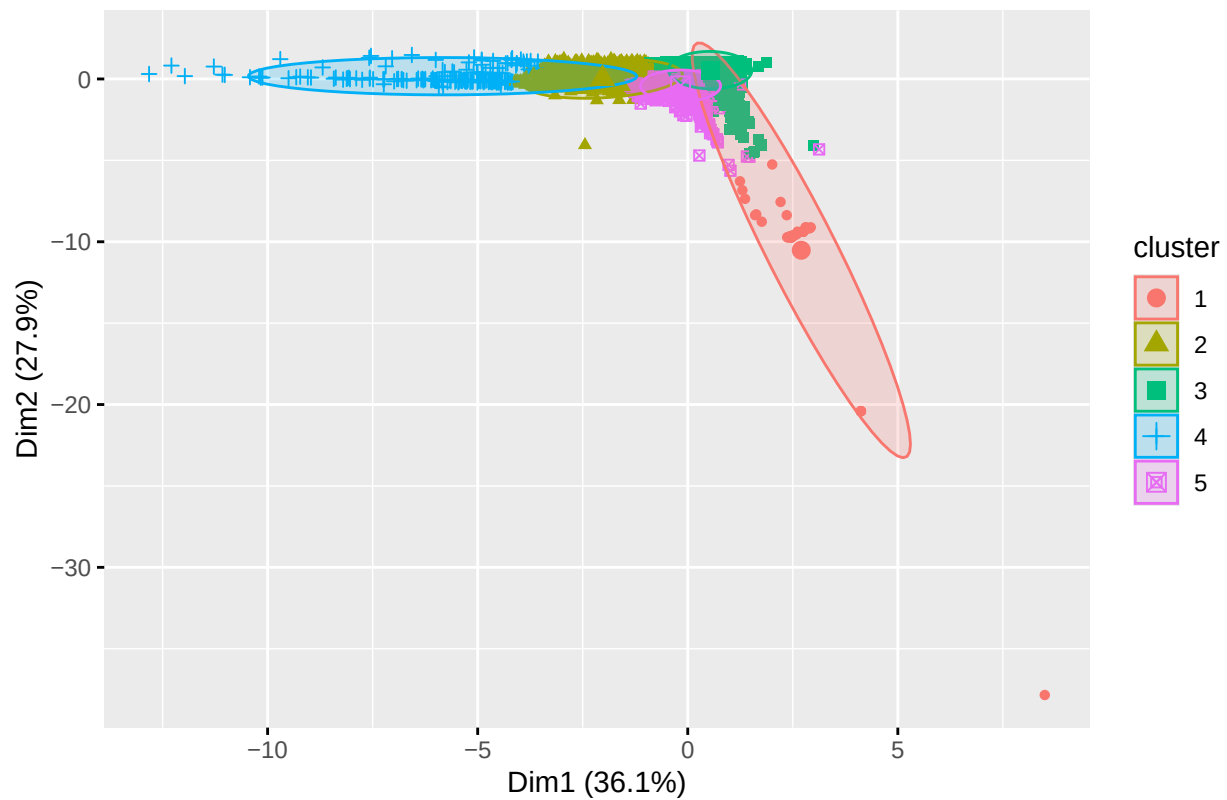
```
km.out3 <- kmeans(airbnb_na, centers = 5, nstart = 20)
```

```
fviz_cluster(km.out3, data = airbnb_na, geom = "point",
             stand = FALSE, frame.type = "norm")
```

```
## Warning: argument frame is deprecated; please use ellipse instead.
```

```
## Warning: argument frame.type is deprecated; please use ellipse.type instead.
```

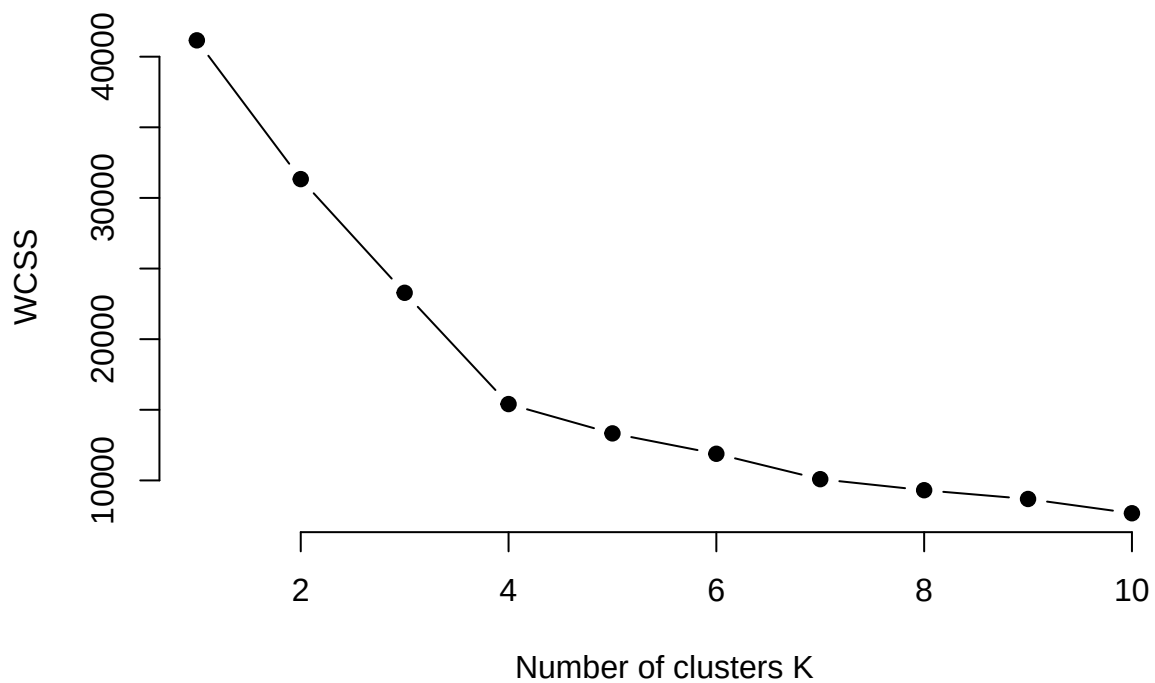
Cluster plot



```
wcss <- function(k) {
  kmeans_model <- kmeans(airbnb_na, centers = k, nstart = 25)
  return(kmeans_model$tot.withinss)
}
```

```
k_values <- 1:10
wcss_values <- sapply(k_values, wcss)
```

```
plot(k_values, wcss_values, type="b", pch=19, frame=FALSE, xlab="Number of clusters K", ylab="WCSS")
```



```
silhouette_scores <- function(k) {  
  kmeans_model <- kmeans(airbnb_na, centers = k, nstart = 25)  
  ss <- silhouette(kmeans_model$cluster, dist(airbnb_na))  
  mean(ss[, 3])  
}  
  
k_values <- 2:10  
silhouette_values <- sapply(k_values, silhouette_scores)  
  
plot(k_values, silhouette_values, type="b", pch=19, frame=FALSE,  
     xlab="Number of Clusters (k)", ylab="Silhouette Score", main="Silhouette Method")
```

